Educational Investment vs. Outcome

Predictive Machine Learning Exploration
by Bryan Beus

Project Links

The PDF is an abbreviated version of the project

The full project can be found on Github:

https://github.com/blue-santa/comparing-training-req-to-salary-exp

My website and blog contain more of my work:

https://bluesanta.io

Problem Identification

The cost of college education tuition is 40 times more expensive than it was in 1963.

Adjusted for inflation, this is an increase of 312.4%

(source : Hanson, Melanie. “College Tuition Inflation Rate” EducationData.org, 2025-11-26, https://educationdata.org/college-tuition-inflation-rate)

Problem Identification

In 1963, the average household income was $6,700/year.

Adjusted for inflation, that’s $62,000/year.

(source: https://www.census.gov/library/publications/1964/demo/p60-043.html)

Problem Identification

In 2026, average household income is around $85,828/year.

Adjusted for inflation, this represents only a 32.3% increase since 1963.

(source: https://www.visualcapitalist.com/mapped-household-income-varies-major-us-metros/)

Problem Identification

A prospective college student may want to carefully consider whether their chosen educational investment will provide a suitable financial return.

Problem Identification

This project, “Comparing Training Requirements to Salary Expectations,” addresses this problem.

Specifically, this study compares the amount of education that is required for a career against its financial outcome.

Our Two Datasets

BLS Wage Data

- Provides wage data
- Careers are joinable on “OCC Codes”

O*NET Career Data

- Provides survey data where respondents indicate how much education is needed for career success
- Careers are also joinable on “OCC Codes”

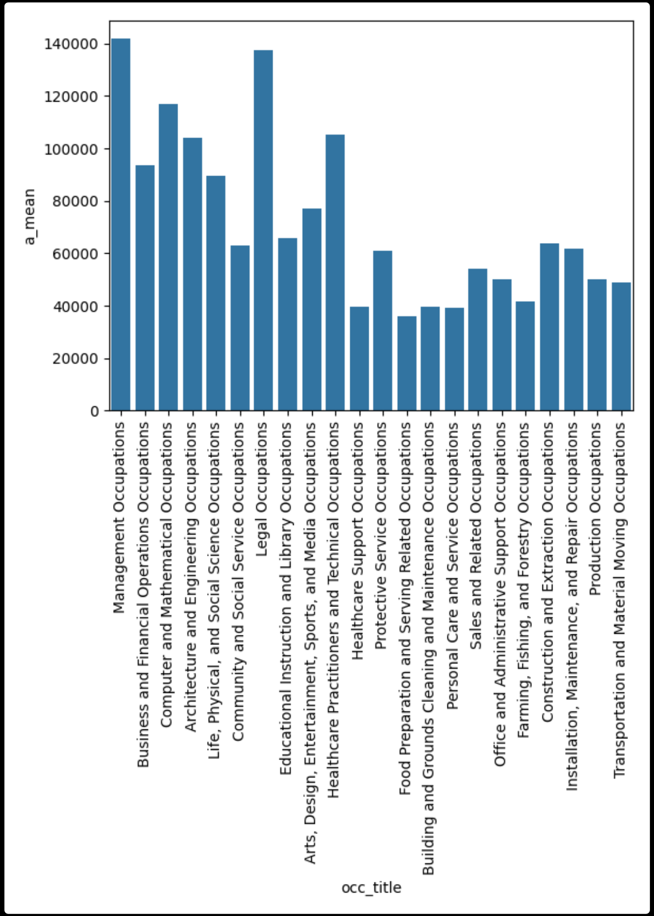
Data Wrangling

Data for each set was downloaded from their respective websites

Each dataset went through several stages of wrangling

A PostgreSQL database provided archival support

Exploratory Data Analysis

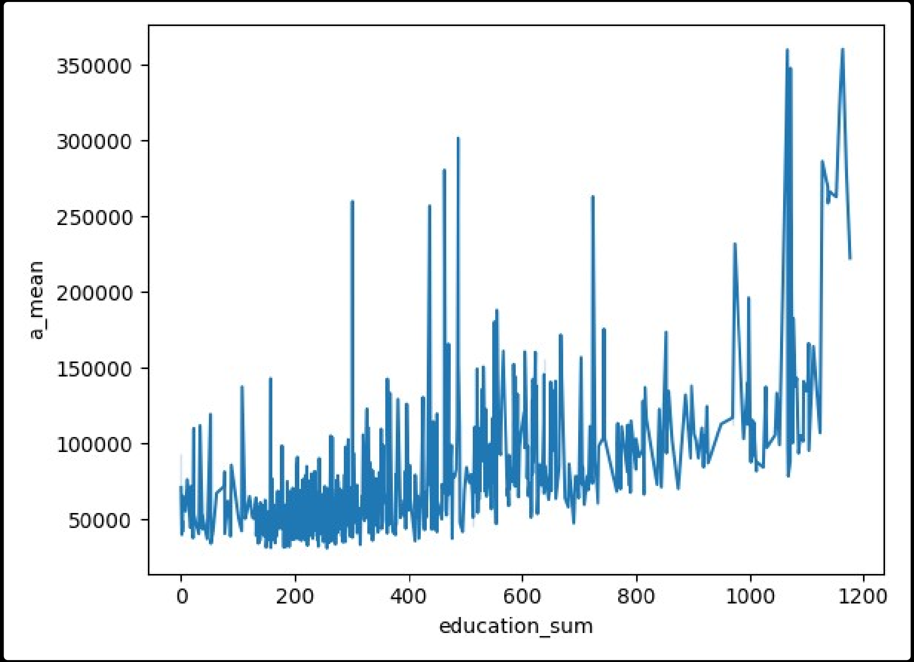


At a glance some careers were noticeably more financially rewarding than others:

Management and Legal occupations were the most lucrative

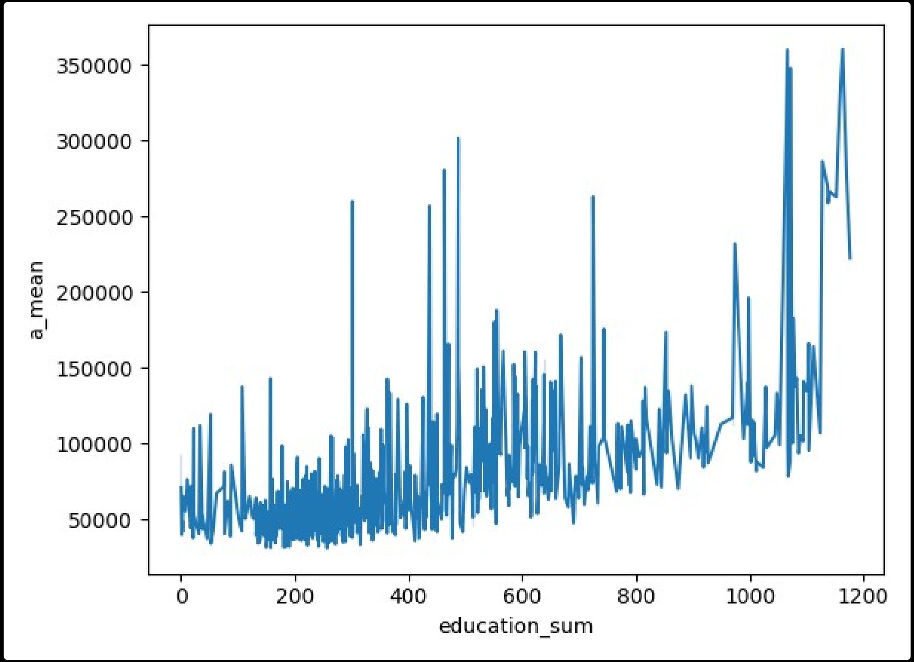
Food Preparation and Serving occupation were the lowest

Exploratory Data Analysis



More education slowly but surely leads to a higher expected income

Exploratory Data Analysis



However, some careers become lucrative with far less educational investment

Lesser Education - Higher Income

Three careers that provided higher levels of income while requiring less educational investment:

- Athletes and Sports Competitors
  - Education Score: 301
  - Average salary: $259,750
- Family Medicine Physicians
  - Education Score: 437
  - Average salary: $256,830
- Airline Pilots, Copilots, and Flight Engineers
  - Education Score: 463
  - Average salary: $280,570

Higher Education - Higher Income

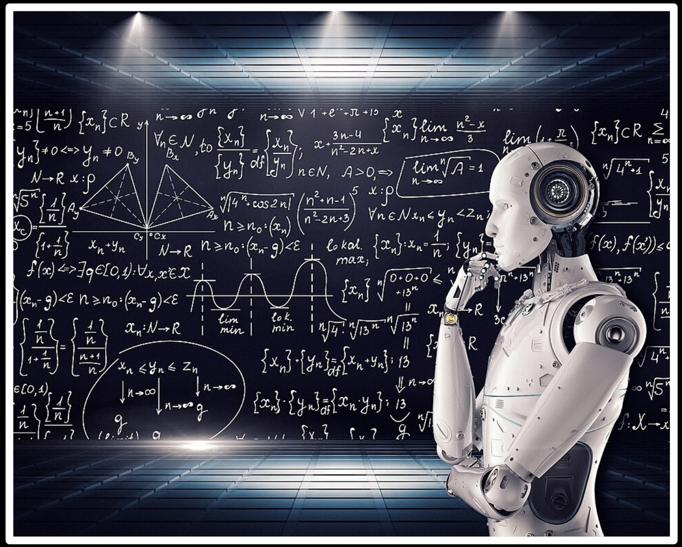
Three careers that provided higher levels of income, but required significant educational investment:

- Oral and Maxillofacial Surgeons
  - Education score: 1164
  - Average income: $360,240
- Obstetricians and Gynecologists
  - Education score: 1171
  - Average income: $281,130
- Pediatricians, General
  - Education score: 1177
  - Average income: $222,340

Predictive Modeling

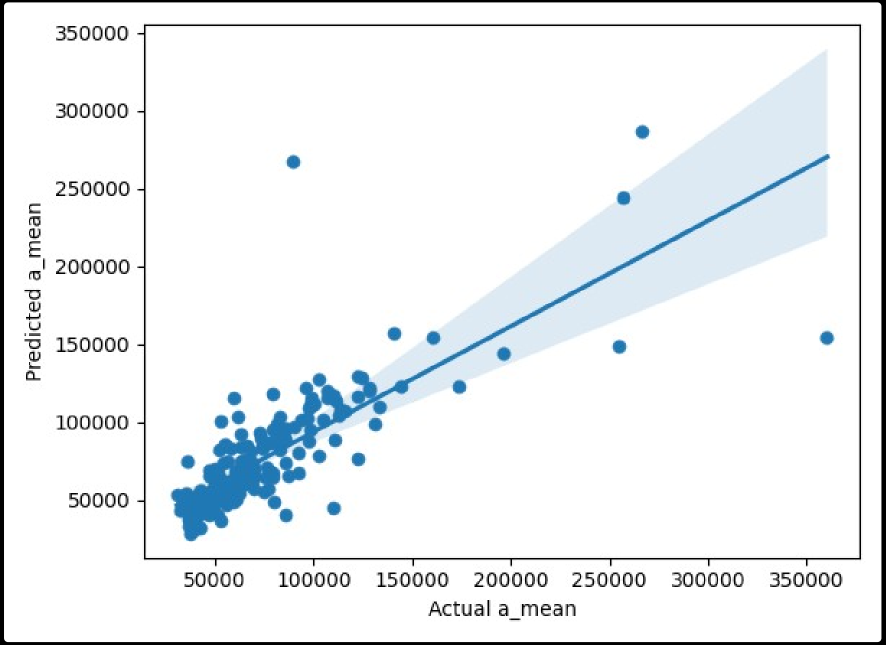
The study explored three different models to predict outcome:

- Linear Regression
- Ordinary Least Squares
- Ridge Regression



Linear Regression

The Linear Regression model had the highest R^2 score at 0.795



Credits

Study conducted by Bryan Beus: bluesanta.io

Datasets: O*NET, BLS Wage Data

- Image authors and sources:
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